08/01/2025

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Aakash
Medical|IIT-JEE|Foundations

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**MM : 720** NCERT Booster Test Series for NEET (OYM)-2024-25_Test-03 **Time : 200 Min.**

**Topics Covered:**
**Physics:** Nuclei, Semiconductor, Experimental Skills, Electric Charges & Fields, Electrostatic Potential and Capacitance
**Chemistry:** Amines, Biomolecules, Principles Related to Practical Chemistry, Some Basic Concepts of Chemistry
**Botany:** Molecular Basis of Inheritance-II (Upto: From translation to DNA fingerprinting), Ecosystem, Biodiversity and Conservation, Cell: The Unit of Life
**Zoology:** Evolution, Biotechnology: Principles and Processes, Biotechnology and Its Applications, Structural Organisation in Animals: Animal Morphology of Frog, Cockroach

**General Instructions :**

1. There are two sections in each subject, i.e. **Section-A & Section-B**. You have to attempt all **35 questions** from Section-A & only **10 questions** from Section-B out of **15.**
2. Each question carries **+4 marks.** For every wrong response, **–1 mark** shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
3. Use blue/black ballpoint pen only to darken the appropriate circle.
4. Mark should be dark and completely fill the circle.
5. Dark only one circle for each entry.
6. Dark the circle in the space provided only.
7. Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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## PHYSICS

### SECTION-A

1. When a selective impurity is doped into an intrinsic semiconductor, the conductivity of semiconductor
   (1) Increases
   (2) Decreases
   (3) Remains same
   (4) Becomes zero

2. Two charges $q_1$ and $q_2$ are $r$ distance apart. The ratio of electrostatic forces acting on them will be
   (1) $q_1 : q_2$
   (2) $q_2 : q_1$
   (3) $q_1^2 : q_2^2$
   (4) 1 : 1

3. Equipotential surface due to uniformly charged thin wire will be
   (1) Spherical
   (2) Planar
   (3) Cylindrical
   (4) Circular

4. Which of the following is not the property of charge?
   (1) Like point charges repel each other
   (2) For an isolated system net charge is conserved
   (3) Specific charge is invariant
   (4) Charge is quantised

5. Holes are majority charge carriers in

(1) Intrinsic semiconductors

(2) Ionic solids

(3) p-type semiconductors

(4) Both (1) & (3)

6. If a body is charged by rubbing it, its weight

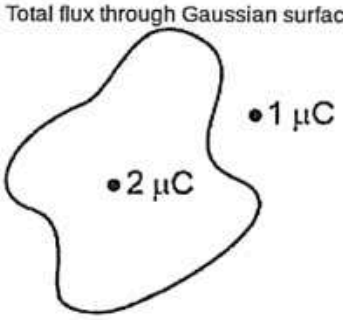
(1) Remain precisely constant

(2) Increases slightly

(3) Decreases slightly

(4) May increase slightly or may decrease slightly

7. Total flux through Gaussian surface shown in figure is




(1) $72\pi \times 10^3\ Nm^2/C$

(2) $36\pi \times 10^3\ Nm^2/C$

(3) $48\pi \times 10^3\ Nm^2/C$

(4) $12\pi \times 10^3\ Nm^2/C$

8. The end correction of a resonance column is 1 cm. If the shortest length resonating with tuning fork is 20 cm, the next resonating length should be

(1) 63 cm

(2) 62 cm

(3) 61 cm

(4) 64 cm

9. When the separation between two point charges is increased, the electric potential energy of the charges

(1) Increases

(2) Decreases

(3) First increases and then decreases

(4) May increase or decrease

10. An electric dipole having dipole moment $\vec{P}$ is placed in a uniform electric field $\vec{E}$. Then the

(1) Potential energy is maximum when $\vec{P}$ is perpendicular to $\vec{E}$

(2) Potential energy is minimum when $\vec{P}$ is perpendicular to $\vec{E}$

(3) Torque experienced by the dipole is maximum when $\vec{P}$ is parallel to $\vec{E}$

(4) Torque experienced by the dipole is maximum when $\vec{P}$ is perpendicular to $\vec{E}$

11. An experiment is performed to find the refractive index of a glass slab using a travelling microscope. In the experiment distances are measured by

(1) A screw gauge on the microscope

(2) A standard laboratory scale

(3) A vernier scale provided on the microscope

(4) A meter scale provided on the microscope

12. A uniform electric field of 40 N/C exists along the *x*-axis in a space. The potential difference ($V_B - V_A$) for the points *A*(4 m, 3 m) and *B*(8 m, 6 m) will be

(1) 120 V

(2) 160 V

(3) –160 V

(4) –120 V

13. A free neutron decays into a proton, an electron and

(1) A neutrino

(2) An antineutrino

(3) An alpha particle

(4) A beta particle

14. A : Nuclear fusion requires very high temperature.
R : For fusion, the light nuclei must have sufficient energy to overcome the coulomb potential barrier.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

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**15.** If the nucleus $^{27}_{13}Al$ has a nuclear radius of about 3.6 fm, then $^{9}_{4}Be$ would have its radius as

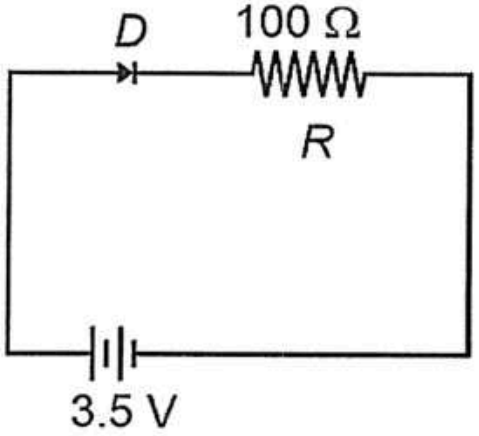
(1) 3.6 fm

(2) $\frac{3.6}{3^{1/2}}$ fm

(3) $\frac{3.6}{3^{2/3}}$ fm

(4) $\frac{3.6}{3^{1/3}}$ fm

**16.** In the given figure, a diode $D$ is connected to an external resistance $R$ = 100 Ω and an e.m.f. of 3.5 V. If the barrier potential developed across the diode is 0.5 V, the current in the circuit will be




(1) 35 mA

(2) 30 mA

(3) 40 mA

(4) 20 mA

**17.** If $n_e$ and $n_h$ are the number of electrons and holes in a semiconductor heavily doped with phosphorus, then

(1) $n_e >> n_h$

(2) $n_e < n_h$

(3) $n_e << n_h$

(4) $n_e = n_h$

**18.** Four points charges $-Q$, $-q$, $2q$ and $2Q$ are placed, one at each corner of the square. The relation between $Q$ and $q$ for which the potential at the centre of the square is zero is

(1) $Q = q$

(2) $Q = -q$

(3) $Q = -\frac{1}{q}$

(4) $Q = \frac{1}{q}$

**19.** Between 2 protons inside the nucleus; the nuclear binding force is $F_1$, coulombic force is $F_2$ and gravitational force is $F_3$, then correct relation is

(1) $F_1 = F_2 = F_3$

(2) $F_1 > F_2 > F_3$

(3) $F_1 > F_2 = F_3$

(4) $F_1 < F_2 < F_3$

**20.** Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R.**
**Assertion A:** Work done in bringing a unit positive test charge from infinity to a point P, located 9 m away from charge $5 \times 10^{-7}$ C, is –500 J.
**Reason R:** This work done is equal to magnitude of electric field at point P due to the charge $5 \times 10^{-7}$ C.
In the light of the above statements, choose the correct answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false.

(4) Both A and R are false

**21.** An isolated parallel-plate capacitor has plate separation $d$ and area of plates $A$. If a slab of dielectric constant 2 of same area but thickness $\frac{d}{2}$ is inserted between the plates, then new capacitance is

(1) $\frac{A\varepsilon_0}{3d}$

(2) $\frac{2A\varepsilon_0}{3d}$

(3) $\frac{4A\varepsilon_0}{3d}$

(4) $\frac{3A\varepsilon_0}{3d}$

**22.** When the jaws of vernier touch each other and zero of vernier scale is to the right of the zero of main scale, then zero error will be

(1) Zero

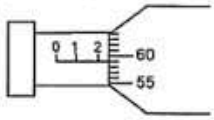
(2) Positive

(3) Negative

(4) Both (2) and (3)

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**23.** The reading of micrometre screw gauge shown in figure (where pitch is 1 mm with 100 divisions on circular scale) is




(1) 2.59 mm

(2) 2.60 mm

(3) 2.61 mm

(4) 3.09 mm

**24.** Given below are two statements:
**Statement I:** The nuclear mass $M$ is always more than the total mass of its constituents.
**Statement II:** The $Q$ -value of a nuclear process is given by $Q$ = (initial kinetic energy – final kinetic energy).
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

**25.** Energy equivalent of 1 kg of substance is

(1) $9 \times 10^{13}$ J

(2) $9 \times 10^{16}$ J

(3) $9 \times 10^{19}$ J

(4) $9 \times 10^{22}$ J

**26.** Electric field due to an electric dipole at a far-away point (whose position vector is $\vec{r}$) on its axis, falls off as:

(1) $r$

(2) $\frac{1}{r}$

(3) $\frac{1}{r^2}$

(4) $\frac{1}{r^3}$

**27.** Match List-I with List-II.

| | List-I (Physical quantity) | | List-II (Corresponding units) |
|---|---|---|---|
| A. | Electric field | I. | V m |
| B. | Electric flux | II. | $V\,m^{-1}$ |
| C. | Dipole moment | III. | $C\,m^{-1}$ |
| D. | Linear charge density | IV. | C m |

Choose the correct answer from the options given below.

(1) A-I, B-II, C-III, D-IV

(2) A-III, B-IV, C-I, D-II

(3) A-II, B-I, C-IV, D-III

(4) A-IV, B-III, C-II, D-I

**28.** A conducting sphere of radius 20 cm has an unknown charge. The electric field due to the sphere at 40 cm from its centre:

(1) May be radially inward

(2) May be radially outward

(3) Must be non-zero

(4) All of these

**29.** Materials with very high resistivity may have energy band gap ($E_g$) equal to

(1) 0.2 eV

(2) 2.2 eV

(3) 1.5 eV

(4) 6 eV

**30.** Consider the following statements A and B identify the correct answer.
A. Diodes can be used for restricting the ac voltage to one direction.
B. By using a capacitor filter in a full-wave rectifier, a constant dc voltage can be obtained.

(1) A is correct but B is incorrect

(2) A is incorrect but B is correct

(3) Both A and B are correct

(4) Both A and B are incorrect

**31.** During the reverse bias of $P$ - $N$ junction, the width of depletion layer

(1) Increases

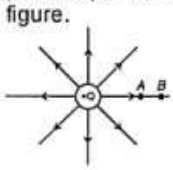
(2) Decreases

(3) Becomes zero

(4) Both (2) and (3)

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**32.** Consider the following electrostatic field lines due to a positive point charge. Two points *A* and *B* are shown in the figure.




The correct statements (about *A* and *B*) among the following are:

I. Potential, $V_A > V_B$.

II. Magnitude of electric field, $E_A > E_B$.

III. Work done by external agency in moving a small negative charge from A to B is positive.

Choose the correct answer from the options given below:

(1) I and II only

(2) III only

(3) II only

(4) I, II and III

**33.** Dimensions of specific heat capacity is

(1) $[ML^2T^{-2}]$

(2) $[M^0L^2T^{-2}K^{-1}]$

(3) $[ML^2T^{-3}]$

(4) $[ML^2T^{-2}K^{-1}]$

**34.** The resistance of a wire depends on

(1) Current flowing through the conductor

(2) Potential difference across the conductor

(3) Dimensions of the conductor

(4) All of these

**35.** Electric field due to a uniformly charged infinite plane sheet (with charge density $\sigma$) at a point *r* distance away is:

(1) $\frac{\sigma}{2\pi\epsilon_0 r}$

(2) $\frac{\sigma}{2\epsilon_0}$

(3) $4\frac{\sigma}{\epsilon_0}$

(4) Zero

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## SECTION-B

**36.** A particle of mass *m* and charge *q* is placed at rest in a uniform electric field *E* and then released. The kinetic energy attained by the particle after moving a distance *y* is

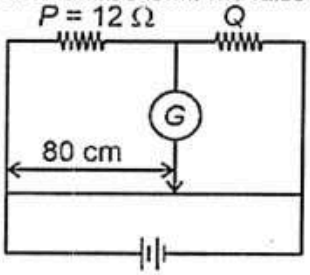
(1) $qEy$

(2) $qEy^2$

(3) $qE^2y$

(4) $q^2Ey$

**37.** In a metre bridge experiment, the null point was found to be at 80 cm as shown. The value of resistance *Q* is




(1) 2 Ω

(2) 3 Ω

(3) 24 Ω

(4) 6 Ω

**38.** The ratio of electron and hole currents in a semi-conductor is $\frac{7}{5}$ and the ratio of drift velocities of electrons and holes is $\frac{7}{4}$, then the ratio of concentrations of electrons and holes will be

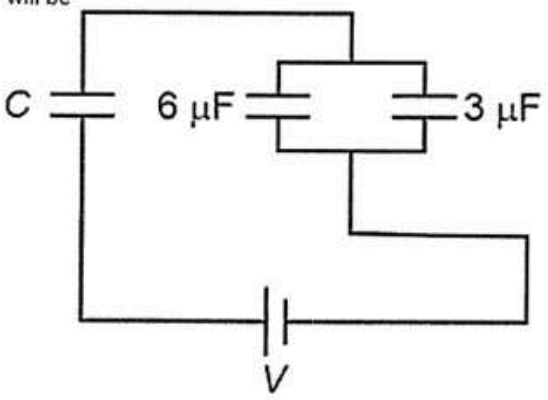
(1) $\frac{5}{4}$

(2) $\frac{4}{7}$

(3) $\frac{4}{5}$

(4) $\frac{5}{8}$

**39.** If charge on 3 μF capacitor in the circuit shown below is 6 μC, then charge on the capacitor of capacitance $C$ (in μC) will be




(1) 6 μC

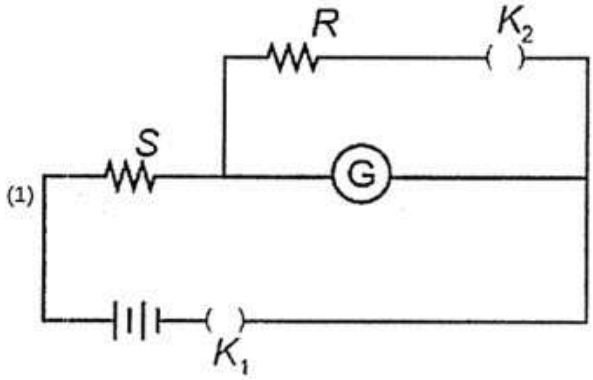
(2) 9 μC

(3) 12 μC

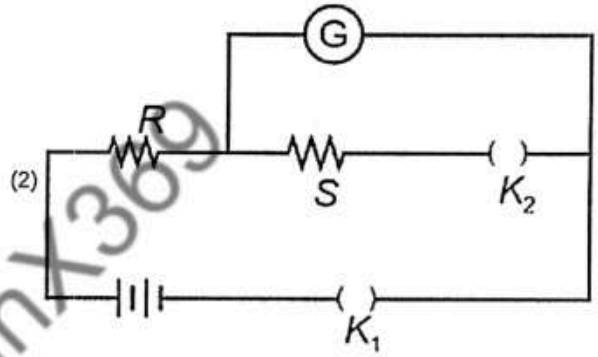
(4) 18 μC

**40.** In the circuit diagrams (1, 2, 3 and 4) shown below, $R$ is high resistance and $S$ is a small resistance connected to galvanometer of resistance $G$. The correct circuit, corresponding to the half deflection method for finding the galvanometer resistance is

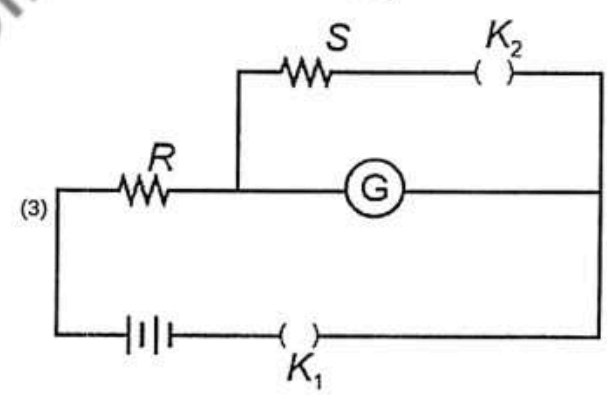
(1)




(2)



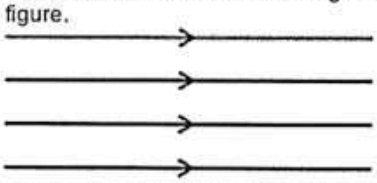

(3)




(4) Both (2) & (3)

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**41.** Electric field lines in a certain region of space are shown in figure.



Then, choose the correct option.

(1) Both magnitude and direction of electric field are uniform

(2) Both magnitude and direction of electric field are non-uniform

(3) Only magnitude of electric field is uniform

(4) Only direction of electric field is uniform

**42.** An electric dipole of dipole moment 1 μC-m is placed along x-axis with its centre at origin. The electric potential at point (0, 4) m due to the given dipole will be

(1) $\frac{9}{16}$ kV

(2) $\frac{9}{8}$ kV

(3) $\frac{2}{3}$ kV

(4) Zero

**43.** A radioactive nucleus ${}^{A}_{Z}X$ decays as shown below, the resultant nucleus (M) will be

${}^{A}_{Z}X \xrightarrow{3\alpha} Y \xrightarrow{3\beta^-} Z \xrightarrow{9\beta^-} M$

(1) ${}^{A-12}_{Z-12}M$

(2) ${}^{A}_{Z}M$

(3) ${}^{A-12}_{Z}M$

(4) ${}^{A}_{Z-12}M$

**44.** If $E_{bn}$ is the binding energy per nucleon and $A$ is mass number, then

A. $E_{bn}$ is lower for light nuclei ($A < 30$)

B. $E_{bn}$ is practically constant for nuclei ($30 < A < 170$)

C. $E_{bn}$ is lower for heavy nuclei ($A > 170$)

D. Nucleons get more tightly bound when nucleus $A = 240$ breaks into two $A = 120$ nuclei.

The correct statements(s) is/are

(1) A and C only

(2) B only

(3) A, B and C only

(4) A, B, C and D

**45.** Electron and positron are:

(1) A particle – antiparticle pair

(2) Identical in mass

(3) Having equal and opposite charges

(4) All of these

**46.** If $10^{19}$ electrons are removed from a negatively charged metal sphere of –1.6 C, then the new charge on the sphere will be

(1) –4.8 C

(2) –3.2 C

(3) Zero

(4) +1.6 C

**47.** The correct statement regarding conductors is:

(1) Electric field at the surface of a charged conductor is $E = \frac{\sigma}{2\varepsilon_0}$, where $\sigma$ is surface charge density

(2) Electric field inside a cavity of any conductor can be non-zero when there is a charge placed in the cavity

(3) The interior of a conductor can have excess charge in the static situation

(4) All of these

**48.** Which of the following dopants, when used in doping the tetravalent Si atom gives a p-type semiconductor?

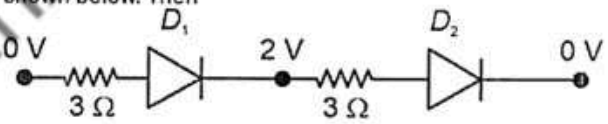
(1) Arsenic

(2) Indium

(3) Antimony

(4) Germanium

**49.** Two *p-n* junction diodes $D_1$ and $D_2$ are connected as shown below. Then




(1) Both $D_1$ and $D_2$ are forward-biased

(2) Both $D_1$ and $D_2$ are reverse- biased

(3) $D_1$ is reverse-biased while $D_2$ is forward-biased

(4) $D_1$ is forward-biased while $D_2$ is reverse-biased

**50.** Choose the correct option for a Zener diode.

(1) It works only in reverse bias

(2) It can maintain constant voltage across the load

(3) Both (1) and (2)

(4) It can 't be used as a voltage regulator

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## CHEMISTRY

### SECTION-A

**51.** The water soluble vitamin among the following is

(1) Vitamin K

(2) Vitamin C

(3) Vitamin D

(4) Vitamin E

**52.** Average atomic mass of $B^{10}$ and $B^{11}$ is 10.81. What is the percentage of $B^{11}$ in the mixture of isotopes?

(1) 10%

(2) 11%

(3) 19%

(4) 81%

**53.** Prussian blue colour observed during detection of nitrogen is due to the formation of

(1) $[Fe(NCS)]^{2+}$

(2) $Fe_4[Fe(CN)_6]_3$

(3) $[Fe(CN)_6]^{3-}$

(4) $[Fe(CN)_5NOS]^{4-}$

**54.** Match groups in list-1 with their group reagent in list-2 and choose correct answer.

| List-1 | | List-2 | |
|---|---|---|---|
| (a) | Group-I | (i) | $NH_4OH$ in presence of $NH_4Cl$ |
| (b) | Group-II | (ii) | $H_2S$ in presence of $NH_4OH$ |
| (c) | Group-III | (iii) | Dilute HCl |
| (d) | Group-IV | (iv) | $H_2S$ gas in presence of dil. HCl |

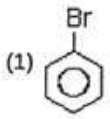
(1) (a)(iii), (b)(iv), (c)(i), (d)(ii)

(2) (a)(i), (b)(ii), (c)(iii), (d)(iv)

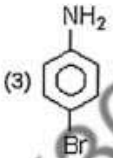
(3) (a)(iv), (b)(iii), (c)(ii), (d)(i)

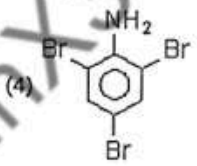
(4) (a)(ii), (b)(i), (c)(iv), (d)(iii)

**55.** Aniline on reaction with $Br_2/H_2O$ majorly gives

(1) 

(2) 

(3) 

(4) 

**56.** A biomolecule contains 0.2% Fe by mass. The minimum molecular mass possible for the biomolecule is (At. Mass of Fe = 56 u)

(1) 14000 u

(2) 28000 u

(3) 5600 u

(4) 2800 u

**57.** Consider the following reaction

$A + 4B + 3C + 2D \rightarrow 5E + 3F$

If 4 mole of A, 14 mole of B, 6 mole of C reacts with 6 mole of D, then the limiting reagent is

(1) A

(2) B

(3) C

(4) D

**58.** Volume of water added to convert 0.5 L aqueous solution of 0.3 M glucose into 0.2 M glucose, is

(1) 0.75 L

(2) 0.5 L

(3) 0.25 L

(4) 0.4 L

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59. Given below are two statements:
**Statement I:** A unit formed by the attachment of a base to 1'-position of sugar is known as nucleoside.
**Statement II:** A nucleotide is formed when nucleoside is linked to phosphoric acid at 5'-position of sugar moiety.
In the light of above statements, choose the most appropriate answer from the options given below.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

60. When dimethyl glyoxime is added to the aqueous solution of nickel chloride in an alkaline medium, it forms a complex having

(1) Black colour

(2) Red colour

(3) Blue colour

(4) Yellow colour

61. Anion that gives smell of rotten egg with warm dil $H_2SO_4$ is

(1) $SO_3^{2-}$

(2) $NO_2^-$

(3) $Cl^-$

(4) $S^{2-}$

62. Layer test is done for detection of

(1) N

(2) S

(3) C

(4) Br

63. Consider the following statements:
**Statement I:** In aqueous solution, amino acids exists as zwitter ions.
**Statement II:** DNA is capable of self duplication during cell division.
In the light of above statements, choose the correct option.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are incorrect

(4) Both statement I and statement II are correct

64. Ratio of crystallization water of potash alum and in Mohr's salt respectively is

(1) 1 : 3

(2) 1 : 2

(3) 2 : 1

(4) 4 : 1

65. Which of the following will give foul smell when reacts with chloroform in the presence of ethanolic KOH

(1) N–methylethanamine

(2) Propan-1-amine

(3) N,N–dimethylmethanamine

(4) 1–nitropropane

66. The base which is absent in DNA is

(1) Adenine

(2) Cytosine

(3) Uracil

(4) Thymine

67. $6.022 \times 10^{22}$ molecules of methane are present in 250 mL of its solution. Molarity of solution is

(1) 2/5

(2) 1/5

(3) 3/5

(4) 6/5

68. On reaction of ferric chloride with NaSCN, blood red colour observed is due to

(1) $Fe(CN)_3$

(2) $Fe(SCN)_3$

(3) $[Fe(CN)_6]^{4-}$

(4) $[Fe(CN)_6]^{3-}$

69. Which among the following amines will not react with Hinsberg's reagent?

(1) $CH_3NH_2$

(2) $C_2H_5NH_2$

(3) $(C_2H_5)_2NH$

(4) $(C_2H_5)_3N$

70. Given below are two statements:
**Statement I:** Globular proteins are soluble in water.
**Statement II:** Keratin and Myosin are examples of globular protein.
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both statement I and statement II are correct

(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct

(4) Both statement I and statement II are incorrect

Einstein3669

71. Number of hydrogen atoms present in two moles of ethane is

(1) $1.2 \times 10^{24}$

(2) $7.2 \times 10^{24}$

(3) $1.08 \times 10^{25}$

(4) $6 \times 10^{23}$

72. Cellulose is a linear polymer of

(1) α-D-Glucose

(2) β-D-Glucose

(3) α-D-Fructose

(4) β-L-Glucose

73. Which among the following ions imparts apple green colour to flame test ?

(1) $Ba^{2+}$

(2) $Sr^{2+}$

(3) $Mg^{2+}$

(4) $Ca^{2+}$

74. Which among the following are essential amino acids?

(1) Alanine

(2) Serine

(3) Threonine

(4) Tyrosine

75. Correct order of boiling point of isomeric amines is

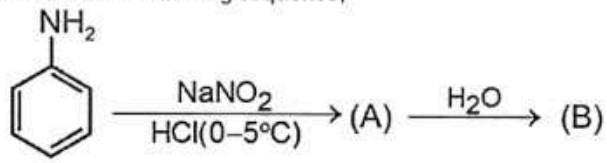
(1) 3° > 1° > 2°

(2) 1° > 2° > 3°

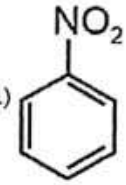
(3) 2° > 1° > 3°

(4) 3° > 2° > 1°

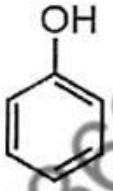
76. Consider the following sequence,

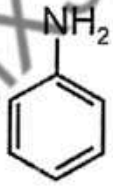



Product (B) is

(1) 


(2) 

(3) 


(4) 


77. Cation which gives brown precipitate with Nessler's reagent?

(1) $Cu^{2+}$

(2) $NH_4^+$

(3) $Mg^{2+}$

(4) $Na^+$

78. Given below are two statements:
**Statement I:** Amylose is long branched chain of α-D(+) glucose units.
**Statement II:** Amylopectin is soluble in water and constitutes to about 80-85% of starch.
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both statement I and statement II are incorrect

(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct

(4) Both statement I and statement II are correct

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79. Number of chiral carbons present in $\alpha$-D Glucose is

(1) 3

(2) 5

(3) 6

(4) 4

80. Number of atoms present in 9 g of $H_2O$

(1) $\frac{3N_A}{2}$

(2) $\frac{6N_A}{2}$

(3) $\frac{N_A}{2}$

(4) 3 $N_A$

81. Maltose is a disaccharide which consists of

(1) $\alpha$-D Glucose and $\beta$-D Fructose

(2) $\alpha$-D Glucose and $\beta$-D Glucose

(3) $\alpha$-D Glucose and $\alpha$-D Glucose

(4) $\beta$-D Galactose and $\beta$-D Glucose

82. Glucose on reaction with $Br_2$ water yields

(1) n-Hexane

(2) Glucaric acid

(3) Gluconic acid

(4) Saccharic acid

83. Number of significant figures is 13.3201500 is

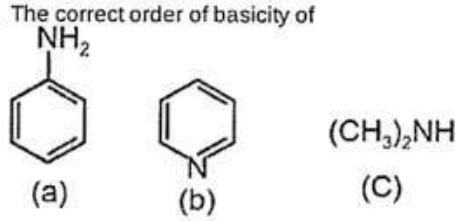
(1) 7

(2) 9

(3) 6

(4) 8

84. The correct order of basicity of




(a) (b) (C)

(1) (a) > (b) > (c)

(2) (b) > (c) > (a)

(3) (c) > (b) > (a)

(4) (c) > (a) > (b)

85. In DNA, adenine forms hydrogen bonds with

(1) Cytosine

(2) Thymine

(3) Guanine

(4) Uracil

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## SECTION-B

86. Match List-I with List-II.

| List I (Concentration terms) | | List II (Units) |
|---|---|---|
| a. Molarity | (i) | mole |
| b. Mole fraction | (ii) | Unitless |
| c. Molality | (iii) | mol $L^{-1}$ |
| d. Amount of substance | (iv) | mol $kg^{-1}$ |

Choose the correct answer from the options given below.

(1) a(iv), b(iii), c(ii), d(i)

(2) a(ii), b(iv), c(i), d(iii)

(3) a(iii), b(ii), c(iv), d(i)

(4) a(iv), b(ii), c(i), d(iii)

87. If the mass ratio of methane and sulphur trioxide is 4 : 1 then the ratio of their number of atoms will be

(1) 4 : 1

(2) 8 : 1

(3) 8 : 5

(4) 25 : 1

88. 49 g $H_2SO_4$ is added in 1500 g of water. The molality of the solution is (mol mass $H_2SO_4$ = 98 u)

(1) 0.24 m

(2) 0.33 m

(3) 0.98 m

(4) 1.33 m

**89.** Given below are two statements one is labelled as Assertion (A) other is labelled as Reason (R).
**Assertion (A):** During nitration of aniline, significant amount of meta derivative is formed.
**Reason (R):** In strongly acidic medium, aniline is protonated to form the anilinium ion which is meta directing.
In the light of the above statements, choose the **correct** answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) (A) is true but (R) is false

(3) (A) is false but (R) is true

(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

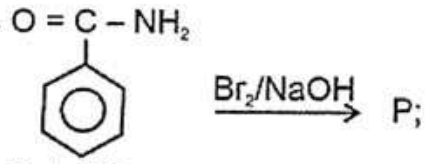
**90.** Given below are the two statements.
**Statement I:** Benzene diazonium chloride is a colourless crystalline solid.
**Statement II:** Benzene diazonium chloride is insoluble in water.
In light of the above statements, choose the correct answer.

(1) Statement I is correct but statement II is incorrect

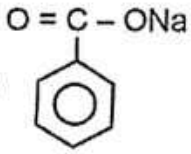
(2) Statement I is incorrect but statement II is correct

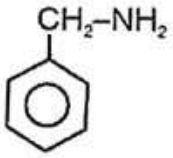
(3) Both statement I and statement II are correct

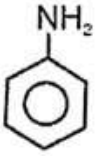
(4) Both statement I and statement II are incorrect

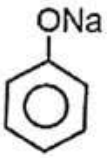
**91.** 


Product P is

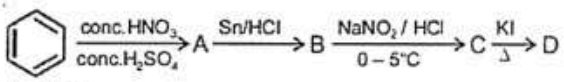
(1) 


(2) 

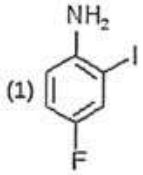

(3) 

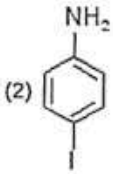

(4) 

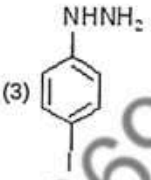

**92.** Consider the following reaction

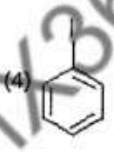



Product D is

(1) 


(2) 


(3) 


(4) 

**93.** Deficiency of which vitamin causes beri beri?

(1) Thiamine

(2) Riboflavin

(3) Pyridoxine

(4) Retinol

**94.** A hydrocarbon consists of carbon and hydrogen in the weight ratio of 12 : 1. What is molecular formula of the compound if its vapour density is 39?

(1) $C_3H_3$

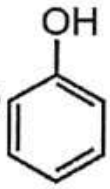
(2) $C_2H_2$

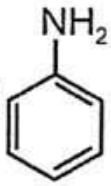
(3) $C_6H_6$

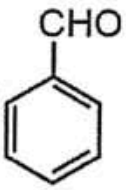
(4) $C_4H_4$

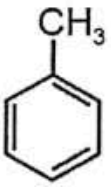
EinsteinX369

**95.** On reaction of (A) with benzene diazonium is chloride acidic medium yields a yellow dye, then A is

(1) 

(2) 

(3) 

(4) 

**96.** Which of the following test cannot be used to distinguish propanone and propanal?

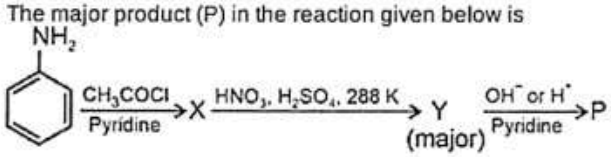
(1) Schiff test

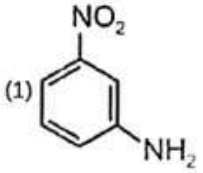
(2) Fehling's test

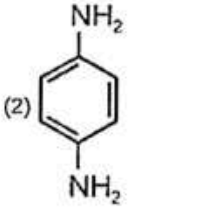
(3) Tollens' test

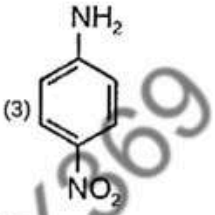
(4) 2,4-DNP test

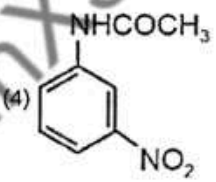
**97.** The major product (P) in the reaction given below is




(1) 

(2) 

(3) 

(4) 

**98.** Given below are two statements.

**Statement I:** $PbSO_4$ is highly soluble in water.

**Statement II:** $PbSO_4$ is soluble in ammonium acetate solution.

In the light of above statements choose the **correct** option.

(1) Both the statements I and II are correct

(2) Both the statements I and II are incorrect

(3) Statement I is incorrect and statement II is correct

(4) Statement I is correct and statement II is incorrect

EinsteinX369

**99.** The oxidation state of Fe in its brown ring complex is

(1) +1

(2) +2

(3) +3

(4) +4

**100.** In Borax bead test, which of the following compound is formed?

(1) Metaborate

(2) Tetraborate

(3) Borazine

(4) Orthoborate

## BOTANY

## SECTION-A

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**101.** Match **column I** with **column II** and select the **correct** option.

| | Column I | | Column II |
|---|---|---|---|
| (a) | Anton von Leeuwenhoek | (i) | Chromatin |
| (b) | Robert Brown | (ii) | Ribosomes |
| (c) | George Palade | (iii) | Nucleus |
| (d) | Flemming | (iv) | Live cells |

(1) a(iii), b(iv), c(i), d(ii)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(iv), b(iii), c(i), d(ii)

(4) a(ii), b(iv), c(iii), d(i)

**102.** State True **(T)** or **False(F)** for the following statements and select the **correct** option.

(A) Anything less than a complete structure of a cell does not ensure independent living.

(B) Plasmid DNA is used to monitor bacterial transformation with foreign DNA.

(C) Larger and more numerous nucleoli are present in cells actively carrying out lipid synthesis.

| | A | B | C |
|---|---|---|---|
| (1) | T | T | T |
| (2) | F | T | F |
| (3) | T | T | F |
| (4) | F | F | T |

(1) 1

(2) 2

(3) 3

(4) 4

**103.** Which of the following organelles is the important site of formation of glycoproteins and glycolipids?

(1) Lysosome

(2) Golgi apparatus

(3) Mitochondrion

(4) Sap vacuole

**104.** Select the **incorrectly** matched pair.

(1) Amyloplasts – Starch

(2) Aleuroplasts – Protein

(3) Elaioplasts – Cellulose

(4) Chromoplast – Fat soluble pigments

**105.** Vertical distribution of different species occupying different level is called

(1) Stratification

(2) Zonation

(3) Divergence

(4) Pyramid

**106.** Read the following statements and select the **correct** option.
**Statement A :** Gross primary productivity of an ecosystem is the rate of production of organic matter during photosynthesis.
**Statement B :** Net primary productivity is the available biomass for the consumption to herbivores and decomposers.

(1) Only statement **A** is correct

(2) Only statement **B** is correct

(3) Both the statements **A** and **B** are correct

(4) Both the statements **A** and **B** are incorrect

**107.** Which of the following statements is **not** true w.r.t. the powerhouses of the cell?

(1) They are the sites of aerobic respiration

(2) Matrix possesses a circular DNA, a few RNA molecules and 40S as smaller subunit of ribosomes

(3) They divide by fission

(4) Their number depends upon the physiological activity of the cell

**108.** Select **odd** one out w.r.t the endomembrane system.

(1) Endoplasmic reticulum

(2) Vacuoles

(3) Golgi bodies

(4) Microbodies

**109.** Read the following Assertion (A) and Reason (R) and choose the **correct** option.
**Assertion (A):** Concentration of ions is significantly higher in the vacuole than in the cytoplasm.
**Reason (R):** Tonoplast facilitates the transport of a number of ions and other materials against the concentration gradient into the vacuole.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) (A) is true but (R) is false

(3) Both (A) and (R) are false

(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

**110.** India having more than 50,000 genetically different strains of rice, and 1000 varieties of mango shows

(1) Species diversity

(2) Community diversity

(3) Genetic diversity

(4) Ecological diversity

**111.** The term 'biodiversity' was popularised by

(1) Edward Wilson

(2) Alexander von Humboldt

(3) David Tilman

(4) Paul Ehrlich

**112.** Match **column I** with **column II** and choose the **correct** option.

| | **Column I** | | **Column II** |
|---|---|---|---|
| a. | Dodo | (i) | Russia |
| b. | Steller's sea cow | (ii) | Mauritius |
| c. | Thylacine | (iii) | Africa |
| d. | Quagga | (iv) | Australia |

(1) a(ii), b(iv), c(iii), d(i)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(ii), b(i), c(iv), d(iii)

(4) a(ii), b(i), c(iii), d(iv)

**113.** Estimates made by Robert May placed the global species diversity at about

(1) 10 million

(2) 7 million

(3) 5 million

(4) 15 million

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**114.** Extinction of steller's sea cow and passenger pigeon occurred due to

(1) Alien species invasion

(2) Endemism

(3) Over-exploitation by humans

(4) Habitat loss and fragmentation

**115.** Read the following statements (A-D).

A. Plants capture only 2-10 percent of the PAR and this small amount of energy sustains the entire living world.

B. There is unidirectional movement of energy towards the higher trophic levels.

C. Humification is the process by which water soluble inorganic nutrients go down into the soil horizon and get precipitated as unavailable salts.

D. Standing crop is measured as the mass or the number of living organisms in a unit area.

Select the option with **correct** ones.

(1) Only A, B and C

(2) Only C and D

(3) Only A, B and D

(4) Only B and D

**116.** The technique of DNA fingerprinting was developed by

(1) H.G Khorana

(2) Griffith

(3) Alec Jeffreys

(4) Oswald Avery

**117.** Read the following statements and select the **correct** option.

**Statement A :** Polymorphism in DNA sequence is the basis of genetic mapping of human genome as well as of DNA fingerprinting.

**Statement B :** Mutations keep on accumulating generation after generation and form one of the basis of polymorphism.

(1) Only statement **A** is correct

(2) Only statement **B** is correct

(3) Both the statements **A** and **B** are correct

(4) Both the statements **A** and **B** are incorrect

**118.** All of the following are the key functional aspects of an ecosystem **except**

(1) Productivity

(2) Decomposition

(3) Energy flow

(4) Stratification

**119.** The rate of biomass production is expressed in terms of

(1) $(\text{kcal m}^{-2})\text{yr}^{-1}$

(2) $(\text{kcal m}^{-1})\text{yr}^{-2}$

(3) $\frac{\text{kcal m}^2}{\text{yr}^2}$

(4) $\text{gm}^{-2}\text{yr}^2$

**120.** If in a food chain, energy available for herbivores is 4000 J, then the amount of energy which gets stored and fixed in the tertiary consumer will maximally be

(1) 40 J

(2) 400 J

(3) 0.4 J

(4) 4 J

**121.** Primary productivity of an area depends on

a. Photosynthetic capacity of plants
b. Plant species inhabiting that area
c. Availability of nutrients
d. Variety of environmental factors

Choose the **correct** option.

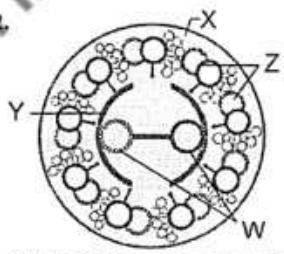
(1) Only a and b

(2) Only c and d

(3) Only b, c and d

(4) All a, b, c and d

**122.**




On the basis of the above figure, identify the **correct** option.

(1) W are connected by an interdoublet bridge

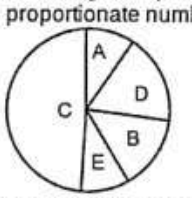
(2) X is mainly composed of lipids and proteins

(3) Y is referred as hub which is a proteinaceous structure

(4) Z are interconnected by the radial spokes

EinsteinX365

**123.** In the given pie chart, the different sectors show the proportionate number of different vertebrate species.




Which sector represents the species of fishes?

(1) B

(2) C

(3) A

(4) D

**124.** Match the **column I** with **column II** and choose the **correct** option.

| | Column I | | Column II |
|---|---|---|---|
| a. | Operator gene | (i) | Interact with the repressor protein |
| b. | Promoter gene | (ii) | Transcribes mRNA for polypeptide synthesis |
| c. | Structural gene | (iii) | Provides attachment site for RNA polymerase |
| d. | Regulator gene | (iv) | Controls the expression of lac *z*, lac *y* and lac *a* genes |

(1) a(i), b(ii), c(iv), d(iii)

(2) a(iii), b(iv), c(ii), d(i)

(3) a(i), b(iii), c(ii), d(iv)

(4) a(ii), b(iv), c(iii), d(i)

**125.** In *lac* operon, the enzyme responsible for hydrolysis of the disaccharide is coded by

(1) *i* gene

(2) *y* gene

(3) *z* gene

(4) *a* gene

**126.** Loss of biodiversity in a region may lead to all, **except**

(1) Decline in plant production

(2) Greater resistance to drought

(3) Increased variability in pest and disease cycle

(4) Increased variability in plant productivity

**127.** Identify the **incorrectly** matched pair.

(1) The Amazon rain forest – Lungs of the planet

(2) Western Ghats of India – Very low level of species richness

(3) Fruit eating birds – Frugivorous

(4) Higher biodiversity – Higher productivity

**128.** Choose the **odd** one w.r.t. humus.

(1) Dark coloured amorphous substance

(2) Colloidal in nature

(3) Undergoes decomposition at very fast rate

(4) Serves as a reservoir of nutrients

**129.** An elaborate network of filamentous proteinaceous structures consisting of microtubules and microfilaments is collectively referred to as the

(1) Thylakoid

(2) Flagella

(3) Cisternae

(4) Cytoskeleton

**130.** The chromosomes in which centromere is situated close to one end are

(1) Sub-metacentric

(2) Metacentric

(3) Acrocentric

(4) Telocentric

**131.** Read the following statements (A-D).

A. The order and sequence of amino acids cannot be defined by the sequence of bases in the mRNA.

B. 23S rRNA in bacteria acts as a catalyst for the formation of peptide bond.

C. The UTRs are present at both 5' end before start codon and at 3' end after stop codon

D. Translation process is an energy independent process.

The **correct** statements are

(1) Only A, B and D

(2) Only B and C

(3) Only A and C

(4) Only B, C and D

Einstein 369

**132.** Read the following Assertion **(A)** and Reason **(R)** and choose the **correct** answer from given options.
**Assertion (A) :** In an aquatic ecosystem, pyramid of biomass can be inverted.
**Reason (R) :** Biomass of zooplanktons is higher than that of phytoplanktons, as the life span in the former is longer and the latter multiply much faster, though having shorter life span.

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

(2) Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**

(3) **(A)** is true but **(R)** is false

(4) Both **(A)** and **(R)** are false

**133.** Identify the **mismatched** pair..

(1) Nerve cell – Branched and long

(2) Tracheid – Elongated

(3) Mesophyll cell – Dumbbell-shaped

(4) Red blood cell – Round and biconcave

**134.** In cyanobacteria, the membranous extensions into the cytoplasm which contain photosynthetic pigments are called

(1) Inclusions bodies

(2) Mesosomes

(3) Chromatophores

(4) Microbodies

**135.** Nucleolus is an active site for the synthesis of

(1) Ribosomes

(2) Ribosomal RNA

(3) Double stranded DNA

(4) Centriole

EinsteinX369

## SECTION-B

**136.** Read the following statements and select the **correct** option.
**Statement A :** Plasmid DNA confers certain unique phenotypic characters to bacteria.
**Statement B :** No organelles, like the ones in eukaryotes, are found in the prokaryotic cells.

(1) Only statement **A** is correct

(2) Only statement **B** is correct

(3) Both the statements **A** and **B** are correct

(4) Both the statements **A** and **B** are incorrect

**137.** An elaborate network in a cell, called cytoskeleton, consists of all the following structures, **except**

(1) Intermediate filaments

(2) Microtubules

(3) Microfilaments

(4) Fimbriae

**138.** State **True (T)** or **False (F)** for the following statements and select the **correct** option.
(A) Satellite DNA normally does not code for any protein, but they form a large portion of human genome.
(B) Human genome project was a mega project that aimed to sequence every base in human genome
(C) The VNTR belongs to a class of satellite DNA referred to as minisatellite.

| | A | B | C |
|---|---|---|---|
| (1) | T | F | T |
| (2) | T | T | T |
| (3) | F | T | F |
| (4) | F | F | T |

(1) 1

(2) 2

(3) 3

(4) 4

**139.** The Human genome project was completed in

(1) 2004

(2) 2003

(3) 1990

(4) 2002

**140.** Sequencing the whole set of genome that contain all coding and non-coding sequence, and later assigning different regions in the sequence with functions is termed as

(1) Sequence annotation

(2) Expressed sequence tags

(3) Bioinformatics

(4) Gene mapping

**141.** Which of the following statements is **incorrect** w.r.t. species-area relationship?

(1) The great German naturalist and geographer observed that within a region, species richness is inversely proportional to the explored area but only upto a limit.

(2) The relation between species richness and area for wide variety of taxa turns out to be a rectangular hyperbola.

(3) For frugivorous birds and mammals in the tropical forests of different continents, the slope is found to be 1.15.

(4) On logarithmic scale, the relationship is a straight line.

**142.** Read the following statements and select the **correct** option.
**Statement A :** The rate of decomposition is controlled only by the chemical composition of detritus.
**Statement B :** Rate of decomposition is faster if detritus is rich in lignin and chitin.

(1) Only statement **A** is correct

(2) Only statement **B** is correct

(3) Both the statements **A** and **B** are correct

(4) Both the statements **A** and **B** are incorrect

**143.** Select the **mismatched** pair.

(1) Colombia — 1400 species of birds

(2) New York — 105 species of birds

(3) Greenland — 560 species of birds

(4) India — More than 1200 species of birds

**144.** Select **odd** one out w.r.t. broadly utilitarian services of biodiversity of ecosystem.

(1) Drugs

(2) Oxygen

(3) Pollination

(4) Aesthetic pleasure

**145.** Which one is **not** a cause of biodiversity loss?

(1) Alien species invasion

(2) Co-extinction

(3) Endemism

(4) Over-exploitation

**146.** Which of the following statements is **not** true?

(1) Besides flagella, pili and fimbriae are also surface structures of the bacteria but do not play a role in motility

(2) Gas vacuoles are found in blue-green and purple and green photosynthetic bacteria

(3) The cell envelope in most bacteria consists of a tightly bound two layered structure, the outer one is made up of glycocalyx and the inner one is cell wall

(4) In addition to the genomic DNA, many bacteria have small circular DNA outside the genomic DNA

**147.** In ecological pyramids, _____ are not given any place, even though they play a vital role in the ecosystem.
Select the **correct** option to fill in the blank.

(1) Phytoplanktons

(2) Sparrows

(3) Saprophytes

(4) Lions

**148.** Exploring molecular, genetic and species level diversity for products of economic importance is called

(1) Biodiversity

(2) Biopiracy

(3) Biotechnology

(4) Bioprospecting

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149. Choose the **odd** one w.r.t chloroplast.

(1) The stroma contains enzymes required for the synthesis of proteins and carbohydrates

(2) Has 50S and 30S subunits of ribosomes

(3) Possesses thylakoids

(4) Inner membrane is relatively more permeable than the outer membrane

150. Match Column I with Column II and choose the **correct** option.

| | Column I | | Column II |
|---|---|---|---|
| a. | First trophic level | (i) | Top carnivores |
| b. | Primary consumer | (ii) | Carnivores |
| c. | Secondary consumer | (iii) | Herbivores |
| d. | Fourth trophic level | (iv) | Autotrophs |

(1) a(iii), b(iv), c(ii), d(i)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(iv), b(i), c(iii), d(ii)

(4) a(ii), b(iv), c(i), d(iii)

EinsteinX369

## ZOOLOGY

### SECTION-A

151. Which of the following is an Indian organisation that makes decisions regarding the safety of introducing a transgenic plant for public services?

(1) NACO

(2) GEAC

(3) EFB

(4) CDRI

152. In 1997, the first transgenic cow produced human ____ enriched milk which was nutritionally a more balanced product for human babies than natural cow-milk.
Select the correct option to fill in the blank.

(1) Alpha-1-antitrypsin (1.4 g/L)

(2) Alpha-lactalbumin (2.4 g/L)

(3) Insulin-like growth factor (2.4 mg/L)

(4) Recombinant insulin (2.4 g/L)

153. In 1990s, an American company got patent rights on Basmati rice (actually derived from Indian farmer's varieties) through the US Patent and Trademark Office. The term that correctly explains the above statement is

(1) Biofortification

(2) Bioethics

(3) Biopiracy

(4) Bioremediation

154. Select the **correct** option to complete the analogy.
Armyworms : Lepidopterans : : _____ : Coleopterans

(1) Flies

(2) Beetles

(3) Tobacco budworms

(4) Mosquitoes

**155.** RNAi involves silencing of a target mRNA in which there is

(1) Inhibition of transcription of target mRNA

(2) Formation of a complementary dsRNA molecule that prevents translation of the target mRNA

(3) Formation of a dsDNA that prevents translation of non-target mRNA

(4) Specific binding of dsDNA onto the target mRNA

**156.** All of the following are examples of evolution by anthropogenic actions, **except**

(1) Creation of different breeds of dogs by selective breeding

(2) Evolution of different shapes of beaks of Darwin's finches

(3) Appearance of antibiotic resistant bacteria

(4) Selection of pesticide resistant pests

**157.** Which of the following steps of RDT is not included in upstream processing?

(1) Fragmentation of source DNA by restriction enzyme for desired gene of interest

(2) Formulation of products with suitable preservatives

(3) Ligation of the gene of interest into a vector

(4) Transferring the recombinant DNA into the host

**158.** Choose the false statement regarding continuous culture system.

(1) Used medium is drained out from one side while fresh medium is added from the other side.

(2) Cells are maintained in their physiologically most active log phase.

(3) Limited supply of nutrients is present as they are only added at the beginning of process

(4) Production of large biomass leads to higher yields of desired protein

**159.** What would be the consequence if a piece of DNA (not containing *ori*) is somehow transferred into a foreign organism without integration within the host genome?

(1) It will be inherited along with the chromosomal DNA in progeny

(2) It will produce less number of copies in the host cells

(3) It will multiply and get inherited along with the host DNA

(4) It will not be able to multiply by itself in the host cells

**160.** While making bacterial cells competent for transformation, they are treated with divalent cations which/that

(1) Create pores in the plasma membrane of the cells so that foreign DNA can easily get inside the cells

(2) • Increase the repulsion between lipid bilayer of the cells and the foreign DNA

(3) Increase the efficiency with which foreign DNA enters the bacteria through pores in their cell wall

(4) Decrease the repulsion of charges between foreign DNA and hydrophobic lipid bilayer by closing the pores of plasma membrane

**161.** All of the following are direct methods of gene transfer, **except**

(1) Micro-injection

(2) Heat shock method

(3) Biolistics

(4) Disarmed *Agrobacterium* mediated transfer

**162.** The enzyme which is termed as the molecular glue, facilitates the formation of

(1) Disulphide bonds during the synthesis of rDNA

(2) Glycosidic bonds between complementary DNA strands

(3) Phosphodiester bond between two adjacent nucleotides of DNA

(4) Sticky ends on the DNA strands

**163.** All of the following are the desirable qualities in ideal cloning vectors used nowadays for genetic engineering, **except**

(1) Facilitate easy linking of foreign DNA

(2) Allow easy selection of recombinants from non-recombinants due to presence of selectable markers

(3) High copy number

(4) Presence of more than one recognition site for a single restriction enzyme

**164.** Choose the incorrect option regarding recombinant human insulin.

(1) Produced in *E.coli* due to universal genetic code

(2) Does not induce unwanted immunological response in humans

(3) Administered to diabetic patients *via* oral route

(4) Is more effective and safer than bovine insulin

**165.** The hypothesis which proposed that "the formation of diverse organic molecules has occurred from inorganic constituents", was experimentally proved by

(1) Oparin of Russia

(2) Haldane of England

(3) Alfred Wallace and Thomas Malthus

(4) S.L. Miller, an American scientist

EinsteinX3669

**166.** Match Column I with Column II.

| | Column I | | Column II |
|---|---|---|---|
| a. | ELISA | (i) | Amplification of minute quantities of DNA |
| b. | PCR | (ii) | Identification of mutated genes after usage of probe |
| c. | Autoradiography | (iii) | Modern method of diagnosis that serves the purpose of early diagnosis |
| d. | Serum analysis | (iv) | Conventional method of diagnosis |

Choose the correct option.

(1) a(iv), b(iii), c(ii), d(i)

(2) a(iii), b(i), c(ii), d(iv)

(3) a(iv), b(i), c(ii), d(iii)

(4) a(ii), b(iv), c(iii), d(i)

**167.** Choose the **correct** statement among the following w.r.t. biological evolution.

(1) The rate of appearance of new forms of an organism is not linked with the life cycle of that organism.

(2) In nature, there is no genetic basis for getting selected to pass on the traits to progeny.

(3) Adaptive ability is inherited and it has a genetic basis.

(4) Nature selects for fitness which is based on the characteristics which are never inherited.

**168.** Which of the following does not result in evolution in a large population?

(1) Frequent mutation of genes

(2) Random mating

(3) Continuous gene migration

(4) Gene flow

**169.** In 1938, a fish caught in South Africa happened to be a Coelacanth which was thought to be extinct. These animals

(1) Were the ancestors of jawless fishes

(2) Evolved into the first amphibians

(3) Laid thick shelled eggs which did not dry up in the sun

(4) Were immediate descendants of salamanders

**170.** **Assertion (A):** Chimpanzees were the direct ancestors of modern human beings.
**Reason (R):** The skull of baby chimpanzee is more like an adult human skull in comparison to the adult chimpanzee skull.
In the light of above statements, select the correct option.

(1) Both (A) and (R) are false

(2) (A) is false but (R) is true

(3) Both (A) and (R) are true and (R) is the correct explanation of (A)

(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

**171.** Plasmids are often used as cloning vectors. They are

(1) ds linear DNA containing *ori*

(2) ds circular DNA without *ori*

(3) ds circular DNA containing *ori*

(4) ss linear DNA without *ori*

**172.** Match column I with column II w.r.t. plants and their direct ancestors.

| | Column I | | Column II (Direct ancestor) |
|---|---|---|---|
| a. | Horsetails | (i) | Progymnosperms |
| b. | Seed ferns | (ii) | Psilophyton |
| c. | Zosterophyllum | (iii) | Chlorophyte-ancestors |
| d. | Bryophytes | (iv) | Tracheophyte-ancestors |

Select the **correct** option.

| | a | b | c | d |
|---|---|---|---|---|
| (1) | (i) | (ii) | (iv) | (iii) |
| (2) | (iv) | (ii) | (i) | (iii) |
| (3) | (ii) | (i) | (iv) | (iii) |
| (4) | (i) | (iv) | (ii) | (iii) |

(1) 1

(2) 2

(3) 3

(4) 4

**173.** If a hypothetical circular cloning vector is completely digested with a restriction endonuclease 'X' for which it contains 2 restriction sites at an equal distance, then how many band(s) of DNA would be seen in the gel after performing gel electrophoresis?

(1) Two

(2) Three

(3) Zero

(4) One

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174. All of the following exhibit divergent evolution, **except**

(1) Thorns of *Bougainvillea* and tendrils of *Cucurbita*

(2) Banded anteater and sugar glider

(3) Sweet potato (root modification) and potato (stem modification)

(4) Heart of platypus and whale

175. Mature insulin and proinsulin differ from each other as former contains

(1) Only 'A' and 'B' peptide chains

(2) Only one interchain disulphide bond

(3) Two intrachain disulphide bonds

(4) 'A', 'B' and 'C' peptide chains

176. In *Periplaneta americana*, the mouthpart that has grinding and incising regions is

(1) Termed as hypopharynx

(2) Unpaired labrum

(3) Paired mandible

(4) Paired labium

177. Presence of all of the following represents differences between male and female of the respective organisms, **except**

(1) Copulatory pad on the $1^{st}$ digit of the hindlimbs in female frog

(2) Anal styles on the 9th abdominal segment in male *Periplaneta americana*

(3) Vocal sacs in male *Rana tigrina*

(4) Participation of uricose glands for excretion in male *Periplaneta americana*

178. In cockroach, structures present at the junction of midgut and hindgut consists of tubules that are lined by glandular and ciliated cells and are primarily involved in

(1) Grinding of food particles

(2) Secretion of digestive juices

(3) Conversion of nitrogenous waste products into uric acid

(4) Storage of food

179. Select the **correct** pair of structures present exclusively in male and female cockroaches respectively.

(1) Mushroom-shaped gland, Anal cerci

(2) Spermathecae, Titillator

(3) Collaterial glands, Common oviduct

(4) Phallic gland, Spermathecae

180. Read the following statements w.r.t. *Periplaneta americana*.
**Statement (A):** Head lies anteriorly at right angles to the longitudinal body axis and is formed by the fusion of six segments.
**Statement (B):** Forewings and hindwings arise from the thoracic segments.
Choose the correct answer.

(1) Both statements (A) and (B) are incorrect

(2) Only statement (A) is correct

(3) Only statement (B) is correct

(4) Both statements (A) and (B) are correct

181. In which of the following scenarios, air enters through the nostrils into the buccal cavity and then enters into lungs in case of frogs?

(1) During aestivation, when they exhibit buccopharyngeal respiration

(2) While the female is laying eggs during which they exhibit only cutaneous respiration

(3) When they take shelter in deep burrows to protect themselves from extreme cold

(4) When they are on land and exhibit pulmonary respiration

182. Select the option which is not true for both the sexes of frog.

(1) Ureters act as urinogenital duct

(2) Cloaca is the common opening for passage of faecal matter, urine and gametes to the exterior

(3) Presence of urinary bladder ventral to the rectum which opens in the cloaca

(4) Presence of primary sex organs near kidneys

183. Choose the **incorrect** match w.r.t. nervous system of frogs.

(1) Forebrain – Olfactory lobes, cerebral hemispheres and diencephalon

(2) Midbrain – Paired optic lobes

(3) Hindbrain – Cerebellum, pons and medulla oblongata

(4) Cranial nerves – Twenty in number

184. Frogs are poikilotherms because

(1) They don't have a single permanent habitat

(2) They lack the capacity to regulate their body temperature

(3) They can camouflage

(4) They exhibit protective coloration called mimicry

185. In frogs, all of the following are not secreted by the walls of stomach, **except**

(1) Bile

(2) Chyme

(3) Pancreatic juices

(4) HCl and gastric juices

EinsteinX369

SECTION-B

**186.** Which of the following is the **incorrect** statement w.r.t. cockroach?

(1) The wings used in flight are transparent and membranous.

(2) A thick and rigid articular membrane joins dorsal tergites and ventral sternites together.

(3) The blood is composed of colourless plasma and haemocytes.

(4) The number of ganglia in thorax and abdomen are 3 and 6 respectively.

**187.** Select the **correct** option regarding vision in cockroach.

(1) Adapted for nocturnal vision with less sensitivity but more resolution

(2) Several ommatidia in eyes help in receiving several images of an object

(3) Sub-oesophageal ganglion supplies nerves to compound eyes

(4) Each eye consists of only a single unit of hexagonal ommatidium

**188.** Match column I with column II w.r.t. structures and the respective segments in which they are present in the body of *Periplaneta americana*.

| | Column I | | Column II |
|---|---|---|---|
| a. | Testes | (i) | $2^{nd}$ - $6^{th}$ abdominal segments of female |
| b. | Mushroom-shaped gland | (ii) | $4^{th}$ - $6^{th}$ abdominal segments of male |
| c. | Ovaries | (iii) | $6^{th}$ abdominal segment of female |
| d. | Spermathecae | (iv) | $6^{th}$ - $7^{th}$ abdominal segments of male |

Choose the correct option.

(1) a(ii), b(iii), c(i), d(iv)

(2) a(ii), b(iv), c(i), d(iii)

(3) a(i), b(ii), c(iii), d(iv)

(4) a(ii), b(i), c(iii), d(iv)

**189.** In frogs, sinus venosus is a triangular structure which

(1) Joins the left atrium of heart

(2) Is a part of lymphatic system

(3) Receives deoxygenated blood through vena cava

(4) Directly opens into conus arteriosus present on the dorsal side of the heart

**190.** The blood in *Rana tigrina*

(1) Carries only nutrients and electrolytes during circulation

(2) Has enucleated erythrocytes containing haemoglobin

(3) Is composed of plasma, RBCs, WBCs and platelets

(4) Lacks proteins but is composed of plasma and haemocytes

**191.** Louis pasteur disproved the theory of spontaneous generation by his experiment which demonstrated that

(1) Life came only from inorganic molecules

(2) Life came out of decaying and rotting matter like straw, mud, *etc.*, in the absence of air

(3) Life comes only from pre-existing life

(4) Spores which were thought to be the unit of life were transferred to the Earth from space

**192.** All of the following structures are analogous structures *i.e.*, they are not anatomically similar but they perform similar functions, **except**

(1) Eyes of the *Octopus* and of mammals

(2) Wings of butterfly and of birds

(3) Brain of frogs and of humans

(4) Flippers of penguins and dolphins

**193.** Theory of evolution given by Darwin and that by Hugo deVries differ in the fact that

(1) deVries concluded about non-heritable variations which cause evolution

(2) Darwinian variations are large and directionless

(3) Darwin believed that mutation caused speciation

(4) deVries hypothesised that mutations are stochastic

**194.** The closest living relative of dinosaurs are direct descendants of

(1) Synapsids

(2) Sauropsids

(3) Thecodonts

(4) Therapsids

**195.** Insertion of gene of interest in recognition site of which of the following will lead to insertional inactivation of *rop* in pBR322?

(1) *Pvu* I

(2) *Pvu* II

(3) *Hind* III

(4) *Sal* I

EinsteinX360

**196.** In an experiment, the gene of interest was successfully ligated at the recognition site of '*Pvu* I' of pBR322. This gene of interest will express bioluminescence upon addition of its substrate after complete transformation in *E.coli*. Which of the following options will correctly represent the non-transformants?

(1) Will die on both ampicillin and tetracycline containing medium

(2) Will show bioluminescence upon addition of substrate

(3) Will survive on ampicillin containing medium but die on tetracycline containing medium

(4) Will survive on tetracycline containing medium and show bioluminescence in the presence of substrate

**197.** Restriction endonucleases do not facilitate/find use in

(1) Cutting of gene of interest during genetic mapping

(2) Digestion of DNA during experiments of biotechnology

(3) Natural genetic recombination during sexual reproduction

(4) Processes prior to gene cloning for obtaining the desired products

**198.** In a single cycle of PCR experiment, if denatured *Taq* polymerase is added, then which of the following steps would get affected?

(1) Denaturation

(2) Annealing

(3) Extension

(4) Elution

**199.** Read the following statements.
**Statement (A) :** Plant tissue culture was only possible after the discovery of DNA ligase.
**Statement (B) :** Golden rice is the product of somatic hybridisation.
Choose the correct option.

(1) Both statements (A) and (B) are correct

(2) Only statement (A) is correct

(3) Only statement (B) is correct

(4) Both statements (A) and (B) are incorrect

**200.** During gene therapy, a functional ADA cDNA is introduced into cultured lymphocytes by using _____ vector.
Select the correct option to fill in the blank.

(1) pBR322

(2) *Agrobacterium*

(3) Disarmed retrovirus

(4) *E.coli*

EinsteinX369